MATH 1342 – Elementary Statistics – Week 11: Inference for One Proportion and Two Proportions

Nov 2 – 4 | 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) • 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5) **2.** Construct and interpret a confidence interval for a population proportion with z* for 90%, 95%, or 99% confidence. (CLO 6) **3.** Carry out a one-proportion z test and conclude in context, and explain why the CI standard error uses $\hat{p}$ while the test uses p_0. (CLO 8)	e-Text: 6.1-CI Confidence Interval for a Proportion e-Text: 6.1-D Distribution of a Proportion e-Text: 6.1-HT Hypothesis Test for a Proportion Interactive Lecture: 6.1 Inference for a Proportion Slides: 6.1 Inference for a Proportion	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)	WileyPLUS StatKey Blackboard
4. Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CLO 5) **5.** Construct and interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CLO 6) **6.** Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CLO 8)	e-Text: 6.3-CI Confidence Interval for a Difference in Proportions e-Text: 6.3-D Distribution of a Difference in Proportions e-Text: 6.3-HT Hypothesis Test for a Difference in Proportions Interactive Lecture: 6.3 Inference for a Difference in Proportions Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 8, 11:59 PM)	WileyPLUS StatKey Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 5 Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.